



Digital revolution and its impact on education systems in developing countries

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Abstract

Across the world the digital era has proven to transform most education processes and systems. Yet along with this new development a challenge has been how most developing countries including Tanzania can benefit from the digital era while avoiding its downsides in education systems. Through the integrative literature review, the article draws on the recent increasing use of digital technologies among education systems in developing countries. While understanding the infrastructural and resource challenges in the region, the paper seeks to address the knowledge gap related to the digital technology in education by pointing out both the problematic areas and the promising approaches to be adapted in the efforts to harness advantages of digital technology in education processes. It is recommended in this paper that the decision to either use a particular strategy in addressing the technology gap or not, needs to be done carefully so that a particular technology for use is not blindly adopted, rather guided by research evidence that demonstrates productive and counter-productive approaches to technology in use.

Keywords Digital era · Education systems · Digital revolution

1 Introduction

The impact of digital era in teaching and learning is rapidly becoming one of the most important and widely discussed phenomena in contemporary education policy all over the world especially in the developing countries (Accilar 2011; Georgsen and Zander 2013). The technological revolution in education sectors shows some promise of providing the developing countries with unique opportunities to reinvent themselves and overcome the current digital related challenges. Most education experts agree that,

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when properly used, the digital technologies hold great promise of improving the teaching and learning processes (Jelfs and Richardson 2013; Miah and Omar 2012). This situation has raised the need for a capable and technology literate community and with the ability to face different challenges in the contemporary digital era. This need has consequently led to a new and strong desire for most developing countries to equip schools with quality digital related learning materials and facilities, human resources, the elements that appear to be necessary for production of a technologically proficient workforce.

The current technological trajectory promises profoundly appear to show fast paced innovation with no braking system. While the digital ride is exciting, it seems to move so quickly that there is no time to think about the possible unintended consequences that come with it especially in education. With such quick digital changes in the world, the developing countries are likely to end up being unable to effectively respond to critical issues such as effective assimilation of digital technologies in education systems. The challenge is therefore about finding ways to appropriately integrate digital technologies in education systems thereby helping learners to navigate the rapidly moving digital development consciously and reflectively. How to meet this challenge effectively would depend on plans available to address the fundamental question about effective integration of the available digital-technologies in our education systems.

The literature reveals that there have been major shifts in state policy to support digital learning in various parts of the world, as more states are pushing towards the implementation of digital instructional materials in most schools in developing countries (Accilar 2011). Legislators and policy makers are seemingly recognizing the benefits of digital learning and the new demand for instructional materials for learners in the twenty-first century via different devices in schools. Despite available opportunities in digital technologies, studies in developing countries reveal that educators and education researchers have not sufficiently addressed the challenge of developing knowledge practices that can trigger meaningful pedagogy through digital technologies (Accilar 2011; Georgsen and Zander 2013; Miah and Omar 2012).

In many cases, students and teachers in developing countries appear to be under pressure by most education systems to directly use digital technologies with very limited opportunity to question the advantages of such digital technologies or reflect on the role of specific technologies in transforming the teaching and learning processes. Despite the digital revolution bubbling in education sectors in developing countries as well as regular policy updates in the region, many schools have taken a stance of sacred prevailing practices that are free from critical evaluation and therefore leading into a failure to develop novel approaches of teaching and learning through new technologies.

This paper considers the aforementioned challenges with a fundamentally different approach of opening up a discussion regarding effective new pedagogies and classroom didactical measures to be adapted so as to appropriately benefit from the digital technologies in teaching and learning processes. It is expected that the discussion in this paper will highlight important guidelines on how various pedagogical procedures in schools are to be shaped for the sake of benefiting from the digital explosion in education systems.

2 Knowledge gap

The use of digital technologies in today's education processes is different compared to the one in the previous decades. Different countries have been in a rally to tap various advantages that come with the new developments of digital technologies in education sector. All along, the digital technologies have shown positive promises in the developed world in the following areas: Encouraging active learning, knowledge construction, inquiry, and exploration on the part of the learners, all of which allow for remote communication as well as data sharing between teachers and/or learners in different physical classroom locations. The challenge however is on how the technology can effectively be used to enable or make more efficient and effective teaching and learning practices. This paper attempts in a way to articulate this concern.

3 The purpose of the study

The purpose of this paper is specifically to examine various ways through which the digital technology can help in supporting today's teaching and learning processes in developing countries. The outputs of this paper include providing the developing countries with policy recommendations about how to appropriately harness the advantages of digital revolution and applying these advantages in: addressing the current pedagogical and educational challenges; and ensuring that the developing countries develop a system-wide digital based education strategies with an overall goal of shaping the regional digital based education agenda. In such a vision, making access to digital devices and broad-band universal is essential towards achieving the desire to harvest the full advantages of digital era in education.

The paper will partly help in answering the following digital related questions: How can we organize online learning courses including MOOCs to build new learning pathways for learners at all levels of education? How to respond to the challenges linked to quality assurance, pedagogical effectiveness and certification using digital technologies? How can the design and organization, delivery of education be optimized through digital technologies?

The target stakeholders in this paper include Ministers of education, ICT industry leaders, science and technology teachers, ICT experts, policy makers and practitioners, representatives from national, government, non-government, international organizations, researchers, representatives of schools and education institutions.

4 Context of the study

The research context for this paper is the developing countries in the digital era. In this paper the world digital era is situated in several different contexts including: The view about digital era being not limited to tablets, kindles, computers, and smart phones but also includes a social answer to network informatics technologies (Cantoni and Tardini 2010). The second view describes the digital era being about its possession of new ways of bringing together, producing horizontal society through digital technology, and knowledge sharing (Accilar 2011). The third view identifies the digital era in terms of

mushrooming alternative educative tools, such as Massive Open Online Courses (MOOCs) (Georgsen and Zander 2013). The final view about this digital era talks about the inhabitants that include learners born roughly between 1980s to date, representing the first generation to grow up with this new technology and characterized by their familiarity and confidence with information and communications technology (ICT) (Cantoni and Tardini 2010). This generation has been given several nicknames to emphasize its affinity and tendency to use digital technology, such as “generation Y” or “millennials”, “net generation”, “digital learners” “digital natives”, “new millennial learners”, “learners of the digital era”, “digital nerds and digital normals” (Cantoni and Tardini 2010; Thirunarayanan et al. 2011).

5 Methodology

The data source employed in this paper is an *Integrative Research Review* as outlined by Whittemore and Knafl (2005) and Torraco (2005). This method involves five stages which include:

- i. *Problem identification.* This involves building an understanding about the topic at hand while focusing on the central aim of the study. In this paper an understanding was built around exploring the impact of various digital technologies among education systems in developing countries
- ii. *Literature search.* This involves a literature search about the topic at hand. The concept of “digital revolution” was used as a keyword for literature search. By focusing on the search, potentially relevant literature were identified and scrutinized to match the gap aimed to be addressed.
- iii. *Data evaluation.* This is a stage of data appraisal whereby diverse studies were evaluated, selected and coded according to two criteria: theoretical and methodological rigour, and data usefulness and relevance.
- iv. *Data analysis.* This is a stage at which the selected studies were examined based on theoretical and methodological rigour, and data usefulness and relevance. Different themes were identified, categorized, compared and contrasted to build up a story about the topic.
- v. *Data presentation.* A synthesis in the form of a descriptive explanation was developed to systematically present a story about the impact of digital revolution in education systems among developing countries.

The studies by Whittemore and Knafl (2005) and Torraco (2005) present the method of *Integrative Research Review* as an evidence-based practice, used in synthesizing investigations available on the given topic and for directing practice based on scientific knowledge. Although it is quite complex and challenging, in combining data of investigations with diverse design, the integrative research review has proven to reduce biases and errors in the process of rigorous approach of data analysis.

In this paper an integrative literature review is used to generate new frameworks and perspectives about the topic at hand. The integrative research review provides a useful framework that guides the process of selecting relevant literature (Souza et al. 2010; Torraco 2005).

The review process in this paper spans a wide range of studies of any design, quantitative or qualitative, empirical and theoretical research-based articles, books, peer reviewed journals, reports and grey literature (e.g., conference website and published proceedings) published from the 2000s to date. The choice of literature from 2000s to date is based the premise that, the digital era was vibrantly renown from 2000s. Data collection involved a systematic search for all reports relevant to the topic. The integrative literature review was chosen due to several advantages including enabling the researcher to evaluate the strength of the scientific evidence, identify gaps in current research, identify the need for future research, bridge between related areas of the topic, identify central issues about the topic, generate a research question, and identify a theoretical or conceptual framework.

The selected literature not only represents a distillation of the researcher's ability to analyze the trend of digital era and its impact in education sector but also shows a synthesis of past empirical or theoretical studies that provide a more comprehensive understanding of a topic at hand. In this paper, the author attempted to interpret not only the content, but also the context of each piece of the studies reviewed. In so doing, the researcher treated the published works as 'field texts' as suggested by Hamilton and Torraco (2013).

6 International trends of digital revolution

The world is currently in the midst of a digital revolution, the third great modern industrial revolution following those of the late 18th and early 20th centuries. Enormous innovations are being witnessed with surprising digital discoveries (Peterson 2012). Multiple changes are being experienced as it has never been before, with entirely new industries and market sectors all around the world seemingly springing up overnight and a much broader variety of information being available to the public. Education sectors in developing countries are being challenged to adapt the changes in a digital era. With new technologies comes a great deal of changes with a tremendous opportunity for being a part of a burgeoning new wave of revolutionizing the education systems in the world. At stake however, is how developing countries can tap the advantage of the revolution in the digital era and use it in improving their education systems.

Looking at how the world is changing, the evidence shows that old solutions no longer work as more people around the world are gaining access to all the technologies of the digital age, a massive transformation is constantly being experienced in all areas of life including education as a result of the digital revolution (Accilar 2011). Such type of transformation relates to the shift from apprenticeship mode of education to universal education that occurred in the nineteenth century as a result of the industrial revolution. The increasing use of digital technologies has been a good influence in the realms of education, personal relationships and the economy (Georgsen and Zander 2013). This trend seems to have begun from the apprenticeship era, whereby most of what people learned originated from outside of school. Thereafter, the universal education helped people to identify and address the learning needs of schools.

The digital developments in this era are characterized with emergence of powerful push-pull digital related factors which have attracted most of the schools in developing

countries, the outcome of which has increased interest in integrating digital technologies in educational processes. Such change of interest has emerged due to the increasingly boring nature of the schooling processes which most of the time alienate both students and teachers from healthy learning environments. This situation makes the exploding and alluring digital world being relevant alternative for improving the today's learning environments (European Commission 2014). The shift towards integration of digital technologies in learning is also necessitated by the majority of children seeming to develop interest in complex video games and adults interacting with simulations all of sorts (Littlejohn et al. 2012). From such people-digital interactions the new digital technologies appear to create learning opportunities that challenge traditional methods of learning in schools and colleges as a result of which learners are now being enabled to: first, pursue learning on their own terms by taking their education out of school into homes, libraries, internet cafes, and workplaces; and second, make their own decisions on what they want to learn, when they want to learn, and how they want to learn (Jelfs and Richardson 2013).

In the context of this review, the digital revolution is identified by four sets of technological emphases each of which are gaining prevalence in developing countries. These include:

- i. A set of technologies that focuses on redefining teaching and learning through the use of various digital related technologies and programs;
- ii. Attempts to reshape the teaching and learning process by changing the materials of schooling and increasing access to technology related learning resources such as laptops, tablets, and smartphone;
- iii. Technological emphases that attempt to improve education through enhancing educators' individual knowledge, skills and attitude; and
- iv. A set of technologies that attempts to improve the classroom learning environment (Gee 2013; McFarlane 2015).

As much as the digital era might seem inescapable today, is still relatively new and fresh even in the arc of human progress (Jelfs and Richardson 2013). In other words even though most countries are striving to keep ahead of the digital environment, most of them seem to have made reasonable strides towards preparing for knowledge resources in the digital era. However it is evident that the application of digital related devices such as cameras, mobile phones, computers and the internet has had remarkable influence on various aspects of people's daily lives (Accilar 2011; Prensky 2012). For example most students of today's schools in developing countries, prefer to use internet and other related technologies in all facets of their daily lives (such as at school, work, and leisure) (Kolikant 2010).

While there have been massive transformational changes that have been experienced in the developed world including: internet developments, mobile technologies, cloud computing as well as the rise of open education resources, such changes cannot be said for many developing countries where access to digital technologies is much more limited (Hilbert 2011; Miah and Omar 2012). This is in conjunction with the need for digital technologies in education for most institutions in developing countries. In most cases the integration of

digital technologies in education for most developing countries has been important for:

- i. Providing every citizen with the knowledge, skills and competences as well as the lifelong learning opportunities required for living and working in technology rich environment;
- ii. Ensuring that communities can exploit the potential benefits of digital technologies to expand access to quality education and
- iii. Enhancing the quality and relevance of learning throughout life.

To achieve all these requirements the educational processes are pushed towards meeting the needs of individual fulfillment and addressing the concerns of sustainable development of knowledge economies (Rapetti 2012). This process includes disregarding the notion of simply digitizing the prevailing education processes, products, outcomes by focusing on rethinking digital education towards complete educational transformation. Additionally, most countries in the world are being challenged to adapt and keep up with the digital related changes in education. This is especially vivid for countries which have been struggling to advance and prosper without success because they are trapped by inefficient and obsolete educational models.

At the moment, it appears that, there has never been more relevant period than now, with digital learning to become an integral part of education (McFarlane 2015). Digital learning opportunities provide significant contribution towards enabling the learners to own their learning process. However this comes with a responsibility to make sure that all digital related technologies are as contextual and culturally relevant as possible to the all citizens. When this situation is appropriately addressed, such technologies are likely to have a good future promise in education.

7 Digital technology in education arena

The literature identifies four enduring features of digital technology in education such as interactivity, symbolic flexibility, interactive with diverse others, and multi sourced nature of information (Accilar 2011; Hull and Nelson 2005). These are:

- i. **Interactivity:** Describes an interaction in the digital era be it face to face interaction or written conversations whereby one asks questions and receives feedback immediately and fast. It includes merged narratives, reports, emails, blogs postings, or social media connections.
- ii. **Multi-Sourced Knowledge and Experience:** Describes a multi sourced nature of digital related technologies. Given the constant and increasing amount of information in the digital world, the need to develop skills for accessing it from various available multi-sources also increases.
- iii. **Symbolic Flexibility:** Describes a multi-dimensional nature of digital technologies—whereby images (moving and still), words, and sounds are available for flexible use in various educational courses of study. With the multimodal nature of digital technologies it is possible not only to make meaningful interaction with symbols but also to develop an easy understanding of the symbols.

- iv. **Interactive with Diverse others:** Describes the nature of the digital technologies and tools being increasingly easy to use, flexible, ubiquitous, and immediate for communication and knowledge acquisition.

In most developing countries, concerted efforts have been made to initiate the adoption of digital technologies in their education systems. Such initiatives indicate a good promise of improving education processes, enhancing the link of educational experiences to the daily lives and developing skills that youths need for securing jobs in the twenty-first century. For example the “Tanzania beyond Tomorrow programme” initiated for the purpose of closing the digital divide and solving the acute shortage of teachers in secondary schools; the Ugandan inter-connectivity programme known as “Uganda School Net” dedicated to extending educational technology throughout Uganda; and the Senegalese initiative in which teachers and students are using computers extensively as information tools. These are some examples of developing countries’ initiatives towards integrating digital technologies in their education systems and are supported by their governments through the Ministries of Education.

8 Why digital technologies in developing countries’ education systems’ are important

Students in the digital age need to be prepared for both college studies and careers in digital technologies, all of which require the ability to implement digital tools, consume and create digital content faultlessly (Georgsen and Zander 2013). According to Jelfs and Richardson (2013) an access to instant information has infiltrated our society and continuously been part of our personal and professional life. Looking at the modern careers from artists to teachers and mechanics to engineers all depend on digital resources for communication and making their businesses successful. In the education arena, digital development seems to be more important for providing deeper learning experiences for students and transforming the teaching and learning process which in turn requires an integration of digital technologies in classroom instruction to best meet the ever-changing individual and students’ needs. Specifically the digital era appears to be useful in education arena in the following ways:

8.1 Providing opportunities for contextualized learning

Most classroom learning processes in developing countries are contextualized by default in that, the classroom models assume a shape of a shared room equipped with books showing an apprentice model of one-to-one learning and onsite knowledge transfer (Georgsen and Zander 2013). With digital technologies a transformation of the current classrooms could be possible with renewed attention being paid to digitally situated or contextualized learning in contrast to mainstream classroom approaches, with immersive technologies such as virtual worlds, augmented reality games, massive multiplayer games, social networking tools, offering highly active, situated, and social learning experiences.

8.2 Transforming learning from tradition models of learning to digital assisted learning

Most of today's schools in developing countries are commonly characterised by the use of books, notes, calendars, and calculators for helping students to outsource parts of their cognitive functions and, consequently, expanding their intellectual resources. In an information society, there is an increasing opportunity to interact with more intelligent technologies (computers, search engines, artificial intelligence, and other education related software) to expand the students' memory (Brown 2000). In other words, learning in the digital era has been transformed into increasingly blended or hybrid forms in which *Face-To-Face (F2F)* and *Peer-To-Peer (P2P)* based instruction are often combined with virtual learning environments through the use of digital tools, such as social media, computer, tablets, and mobile devices. Learning spaces have also been changed by combining the strength of innovative pedagogical methods and novel digital based tools of learning, thereby creating dynamic and conducive learning environment.

8.3 Providing support for social learning

Most classrooms in developing countries still use traditional classroom models in which each student is regarded as an independent and separate entity with individually owned desk, books, assignments, motivations, assessment, and grades (Miah and Omar 2012). The students' learning progresses are usually evaluated in light of the records of individual student's achievements or failures (Griffin and Care 2015). With digital technologies it has been possible to offer new ways for students to engage in group work, a social learning environment whereby students are better prepared for the networked, globalized market place than those who do not.

8.4 Transforming didactic approaches to constructivist learning

Experiences from the recent classrooms in developing countries indicate that, the teaching process is still based on distribution of text-based materials supplemented with oral information; lessons are reinforced through note taking, homework, and textbook guides; knowledge is possessed by the educator and imparted to his or her students in a top-down, unidirectional transfer; and a student's classroom success or failure is assessed by the said educator (or by an externally mandated examination board) (Miah and Omar 2012). Digital technology has therefore redefined the existing pedagogical roles by eliciting more engagement and investment on the part of the learner, and putting less overt control of knowledge dissemination on the part of the educator.

8.5 Providing diverse learning alternatives for varied students' learning styles

Studies of most classrooms in developing countries reveal the prevalence of traditional learning styles which employ a mechanical model of "one curriculum fits all" under the guise of fairness and efficiency (Georgsen and Zander 2013). The digital era provides different options or pathways towards understanding textual, visual, game based methods of learning (Jelfs and Richardson 2013). The digital development differs from the "*uniform system of learning*" in that the

later is unfair to those students with different learning styles or mental abilities. Contrary to traditional models of learning, the digital era allows for more customized learning within a formal learning context.

8.6 Providing sophisticated tools for learning

The common traditional tools of learning in developing countries have always been books and teachers. With digital era more sophisticated learning support tools for maintaining records of students' learning such as; their performances, and inventory list of supplies, tools for cost accounting, and printing academic reports. The digital technologies are increasingly showing a lot of promises being witnessed in schools whereby many students either spontaneously or with school support are being supported to utilize social media to share co-constructed information regarding their school tasks, for instance, by creating Facebook or WhatsApp groups to support school work.

Therefore students and teachers need to join efforts and function as a comprehensive knowledge-creating community, if they are to tap the full advantage of the digital era and prepared for the emerging innovation-driven knowledge society.

9 Issues about digital technologies to be addressed in education processes

Despite the promising opportunities created by the digital era in the modern schooling, studies by Brown (2000), Miah and Omar (2012), Georgsen and Zander (2013), and McFarlane (2015) identify some issues of concern regarding the digital explosion in developing countries and in other parts of the world with similar contexts. These issues are summarized in Table 1 that follows.

Table 1 Issues facing the successful digitalization of education in developing countries

Economic related issues	Political related issues	Concerns related to cross-cutting issues
Weak, unsupportive and limited digital Infrastructure	Poor government's commitment in investing on integration of digital technologies in education processes	Lack of human resource capital to stir the efforts to integrate ICT in education practices
Infrastructural market-domination by western countries	Less commitment by the international community to assist developing countries adopt/utilize digital technologies	Lack of competencies related to the evolution of digital technologies
Excessive running cost for the digital related programmes	Inappropriate level of regulation(s) to facilitate uptake of digital technologies in education processes	Poor networked information sharing between and within regions
Limited access to digital technologies	Debate between the government and private sector about who takes the leadership role.	Little distribution of new technologies from developed countries to the developing countries
Inadequate digital facilities.	Social and legal constraints leading to censorship and denial of access to certain digital technologies	Poor initiatives towards invention of new technologies

The development of digital technology in today's schooling is also blamed for building classes in education sector in developing countries in that, education which was once viewed as a public good with equal access for all is now up for sale to those who can afford for such specialized services and digitized educational programs (Georgsen and Zander 2013). The other imperative challenge is whether modern schools are capable of coping with challenges from the technology-driven learning practices. Following these challenges outlined in Table 1 the subsection that follows describes the promising measures which according to the literature seem to provide a positive promise in addressing the technology based challenges in education systems among developing countries.

10 Measures to overcome the issues related to digital technology in developing countries' education systems

The efforts to address the challenges of digital technologies in developing countries are not uniform; there are many variables for school, districts, regions, and country leaders to consider as they plan towards enhancing the digital technology use in education environment (Georgsen and Zander 2013). The need to address the challenges of digital technology is built on the premise that, there are numerous prospects for using digital technologies in teaching and learning processes in developing countries. The studies by Scott (2012), Griffin and Care (2015), McFarlane (2015), O'Mara et al. (2015) identify several measures which the governments, International Organizations, development agencies and Aid Banks could adapt in addressing the concerns of the digital era. These include:

10.1 Identifying key scenarios and enablers of digitally enhanced schooling processes

This measure deals with envisioning digital learning through several key players in education. These include; the digital technology industry leaders and policy makers who could specifically be involved in determining the variety of available digital technologies and how such technologies would be supportably structured to achieve the education targets. Under this target all debates need to be anchored at questioning about digital technology's role in ensuring that there is positive impact in education processes.

10.2 Making learning more interactive and connected to the learners' individual needs and goals through various digital means

This measure deals with aligning digital technology programmes so that they provide system-wide support for teachers' effective pedagogical use of digital technologies, therefore optimizing their benefits for effective learning. This could be effectively done by transforming learning to mainstream innovative pedagogies through incentivizing and recognizing teacher innovation in using digital technologies.

10.3 Improving the impact of educators through effective interaction with students

This theme pays attention to the social and human impact of digital education, especially in redefining the role of teachers. The theme describes how digital tools can effectively be used to guarantee more personal and effective interaction with students. This could be done by building multiple lifelong learning pathways that span both formal and informal systems and support job-related skill development. This could also help in finding a way in which a combination of technologies, such as mobile technologies, cloud solutions, and localized content can be leveraged to enhance the quality of teaching and learning processes.

10.4 Addressing the digital skill gap in a rapid changing society

This theme aims at identifying issues that teachers, school leaders and other education managers need to address in their efforts to meet the digital expectations of learners, parents and the society in general. This could be done by introducing new technologies in schools, assessing what needs to be done to improve the pedagogical practices, a situation which is likely to shape and regulate the use of digital devices inside and outside schools.

10.5 Redefining the regulation and policies related to digital technology

This theme aims at redefining regulations and digital policies so that they do not stymie long-term education growth and revolution. The desire is focused at strengthening efforts of adapting approaches that draw more people into the use of internet and other related digital technologies. By redefining policies, it is possible to regulate the connectivity taxes and licensing fees (Miah and Omar 2012). High connectivity taxes and fees appear to not only discourage the growth of digital related services and the economic benefits they would offer but also makes it difficult for firms that provide digital services to make the necessary investments when cost structures are too high. As such, governments need to lower the cost of market access to internet and other digital technologies.

The other general aspects that need to be done in realizing positive outcomes for the digital era include the following:

- a. Developing a strategic plan that would assist the developing countries identify and adopt promising practices for optimal digitization of education systems;
- b. Increase investment in basic digital education related facilities and services in schools and teacher training;
- c. Use existing bilateral relationships and linkages with other developed nations to identify and acquire appropriate technologies in developing countries' education systems;
- d. Developing broader community awareness about the benefits of digital technologies; including the overall contribution to the economic growth as well as the well-being;
- e. Shaping an inclusive transition towards promising and fair digital technology use in all processes of schooling;

- f. Upskilling the workforce (including self-employed workers) by upgrading initial and continuous occupational training programs for the digital era
- g. Ensuring that access to digital education and lifelong learning is good and fair for everyone;
- h. Incorporating the educational perspective in all digital initiatives so that the digital agenda becomes a driving force towards quality education;
- i. Building inclusive and accessible digital public services by ensuring that it supports teachers, school administrators, and learners in running the school activities; and
- j. Developing a framework to safeguard the rights of digital users in online environments as well as establishing fair rules that give equal access to social security and protection of the intellectual property.

11 Conclusion

The paper has articulated the knowledge gap related to digital technology in education but also pointed out both promising approaches and problematic areas to be adopted or addressed respectively in the efforts to harness advantages of digital technology in education processes. The decision to either use a particular strategy in addressing the technology gap or not, needs to be done carefully so that technology use is not blindly adopted, rather guided by research evidence that demonstrates productive approaches. It has been argued in this paper that all digital learning related technologies need to be oriented towards improving learning experiences and outcomes.

For developing countries, the time has come for governments to become champions of digital transformation in their education systems and lead by example. The sooner the governments embark on a digital transformation, the more technology literate individuals are likely to be in making the best of this opportunity. Experiences have shown that there is no way a certain country can avoid the impact of digital era in its education systems. As such it is better to adopt a famous Chinese proverb which says that “*What you cannot avoid, welcome it*”- It does not matter how slowly this can be done, so long as these countries are ready to take full advantage of the era without stopping. The readiness to accept the digital technologies in education systems might require an overall change of methods for educational delivery to some countries. However for this to happen there is a need for all education stakeholders to be prepared and ready to embrace the change.

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